Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking

Team 1 | April 2026

Team Assignments

Team Member	Role & Assignment
Sairam M (Team Lead)	Project oversight, Create records in the Sheet, Conclusion
Karpaga Vinayagam S	Creating Import Table (Staging), Match fields using Mapping Assist
Manuneethi Cholan S	Create Transform Map Configuration
Saravanan R	Transform the Data, Result validation (Testing)
Rada Krishnan S	UI Configuration: Display department to user (Dot-walking)

Executive Summary

- Streamlined onboarding and updates of employee records within ServiceNow via automated data ingestion.
- Utilized native **Import Sets** and **Transform Maps** to transfer external spreadsheet data securely.
- Integrated directly into the core sys_user table, maintaining critical data architecture.
- Enhanced IT Service Desk operational efficiency by implementing dot-walking on the Incident form.
- Significantly reduced manual data entry errors and provided immediate visibility into user context during incidents.

System Context & Infrastructure

System of Record

ServiceNow operates as the centralized system of record for critical enterprise IT operations and management workflows.

Manual Bottlenecks

Historically, manual user creation and baseline updates have proven to be highly inefficient and prone to human error.

Dot-Walking Power

A powerful ServiceNow feature allowing forms to display fields from referenced tables, removing the need for custom scripting.

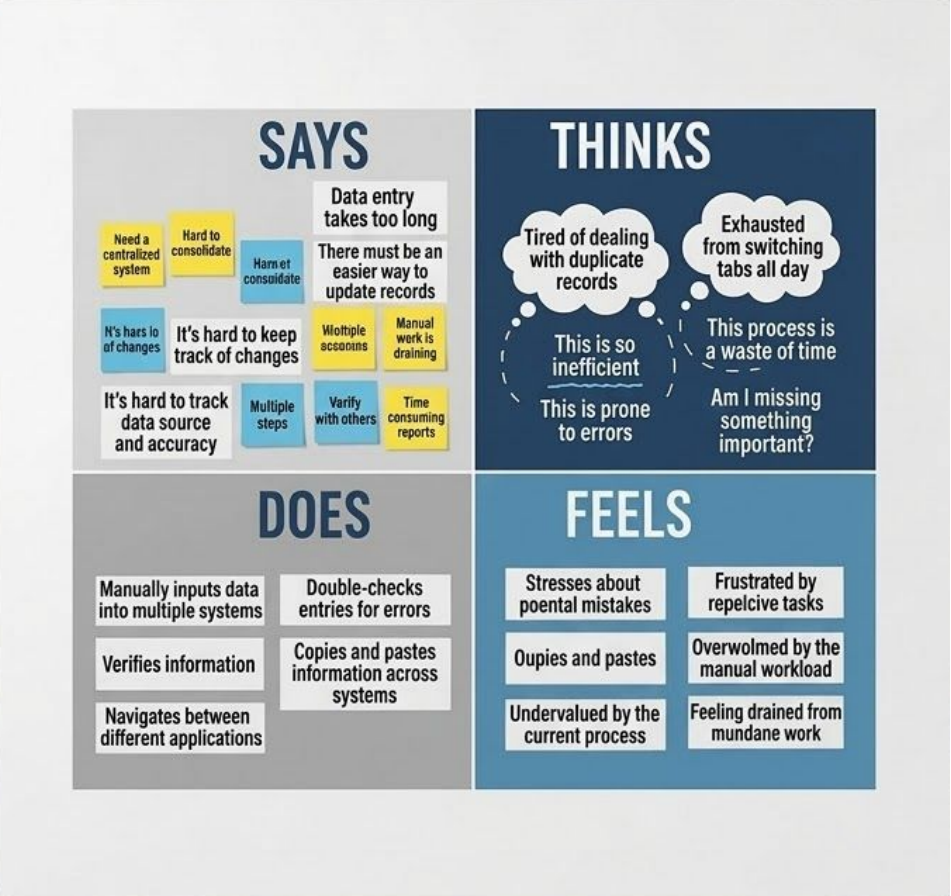
Key User Pain Points

- **Administrators:** Wasting substantial administrative time manually inputting hundreds of employee records during onboarding cycles.
- **Service Desk Agents:** Experiencing inefficiency caused by constantly switching tabs away from the Incident form to look up an Assigned User's details.
- **Data Integrity:** Compromising database health with duplicate user records created due to a lack of automated coalescing and matching checks.

Understanding Empathy

Before deployment, IT Support Agents and System Administrators faced immense friction. The daily operations were hampered by tedious manual entry and navigation overhead.

By mapping out the thoughts, feelings, and actions of these critical users, we identified that the core frustration was the disconnection between data entry systems and active incident resolution workspaces.



Implementation Workflow

1. Data Prep

Format Google Sheet data and export locally as CSV/XLSX file.

2. Stage & Load

Utilize System Import Sets to create the staging table u_import_employee.

3. Transform Map

Map identical fields and set email to coalesce targeting sys_user.

4. UI Config

Apply Form Design to dot-walk user context to Incident workspace.

Objectives & Prerequisites

Project Objectives

1. Establish a repeatable process to ingest CSV/XLSX employee data into the sys_user table.

2. Ensure no duplicate records are created by establishing a unique identifier (Coalesce).

3. Enrich the Incident Management view by surfacing related user data instantly.

System Prerequisites

Access to a ServiceNow Instance with System Administrator (admin) privileges.

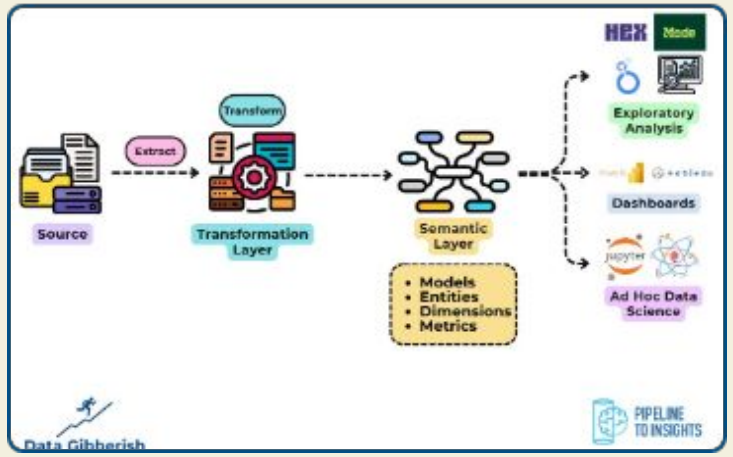
A properly formatted sample spreadsheet populated with test employee records.

Basic understanding of ServiceNow reference fields and table structures.

Core Implementation Phases

Data Loading

Securely staging spreadsheet data into a temporary import set table.



Transformation

Mapping fields and establishing coalesce rules to maintain integrity.

Dot-Walking UI

Configuring the Incident form to dynamically pull related profile details.

Final Project Results

100%

Records Successfully Ingested

Seamless Execution

The Import Set and Transform Map executed seamlessly, proving the integration provides a scalable, reusable pipeline for bulk operations.

Team 1 significantly reduced administrative overhead, eliminated duplicates via coalesce, and greatly enhanced the agent experience by delivering immediate user context on the Incident form.